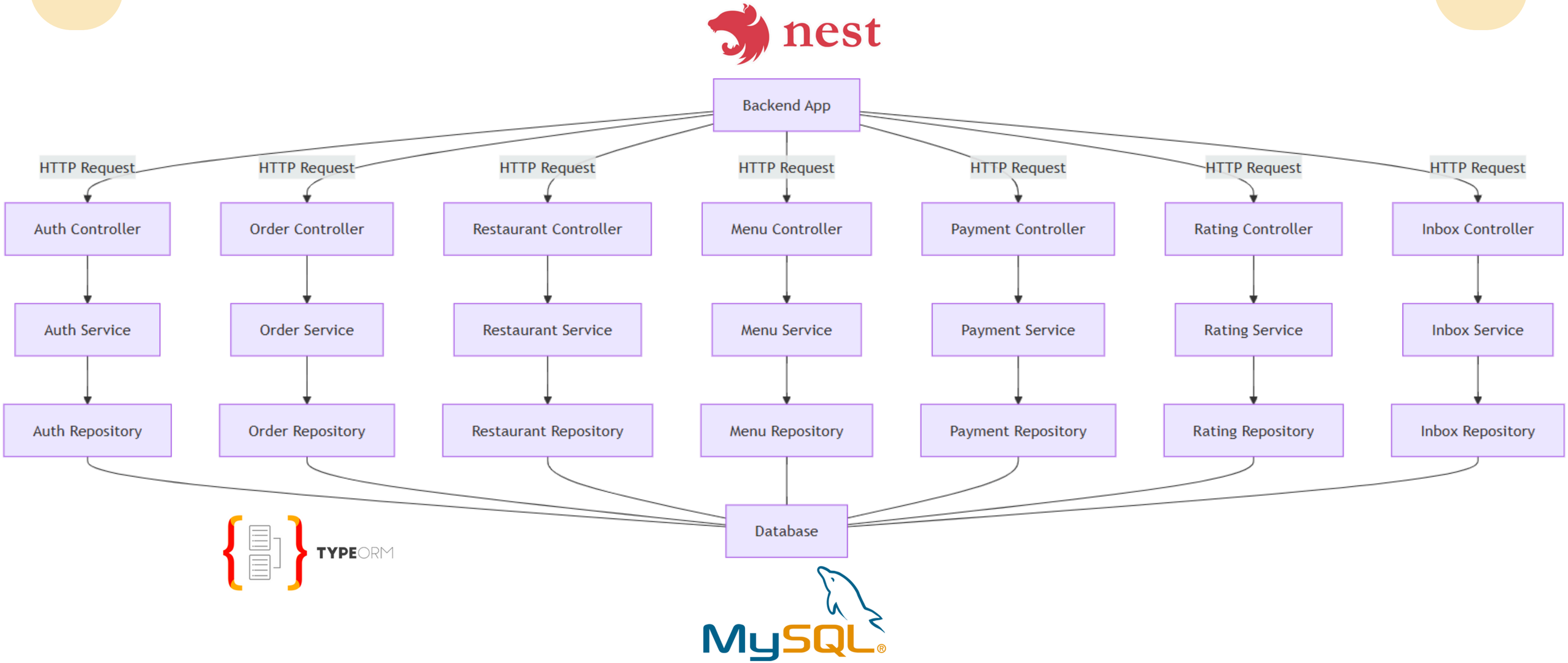
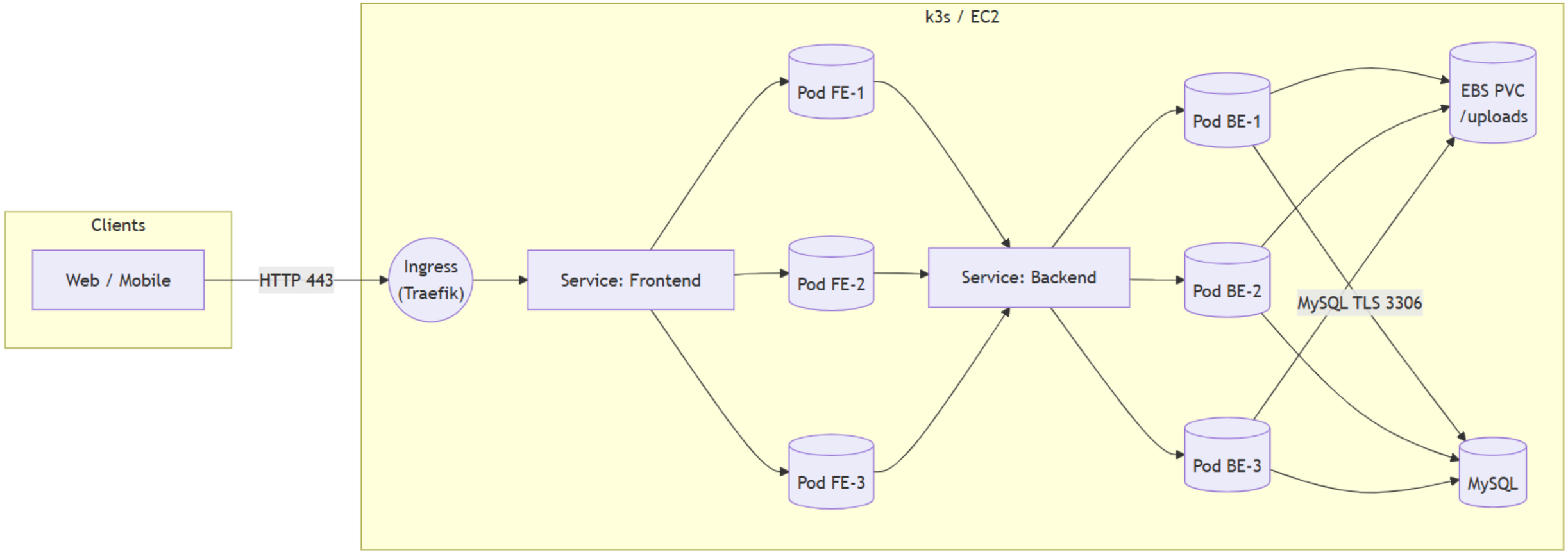
CLOUD NATIVE FINAL MEAL PROVIDER

組員:曾紹幃、李承晏、曾煥宗

後端軟體架構設計


nest
Backend App
HTTP Request
HTTP Request
HTTP Request
HTTP Request
HTTP Request
HTTP Request
HTTP Request
Auth Controller
Order Controller
Restaurant Controller
Menu Controller
Payment Controller
Rating Controller
Inbox Controller
Auth Service
Order Service
Restaurant Service
Menu Service
Payment Service
Rating Service
Inbox Service
Auth Repository
Order Repository
Restaurant Repository
Menu Repository
Payment Repository
Rating Repository
Inbox Repository
Database
TYPEORM
MySQL

系統架構設計


k3s / EC2
Clients
Web / Mobile
HTTP 443
Ingress (Traefik)
Service: Frontend
Pod FE-1
Pod FE-2
Pod FE-3
Service: Backend
Pod BE-1
Pod BE-2
Pod BE-3
EBS PVC /uploads
MySQL TLS 3306
MySQL

USER STORY-員工

- 作為員工，我希望能夠瀏覽餐廳提供的菜單並選擇餐點下單，以便方便地訂購每日餐點。
- 作為員工，我希望下單後能確認訂單明細，以便確保我的訂單正確無誤。
- 作為員工，我希望在用餐後可以對餐點給予星級或評論，以便提供回饋讓其他同事和餐廳了解菜色品質。
- 作為員工，我希望能查看過去的點餐記錄與消費明細，以便追蹤自己的用餐情況或費用報銷。

USER STORY-管理員

- 作為餐廳管理員，我希望能夠新增、修改或下架菜單項目，以便定期調整菜單內容。

- 作為餐廳管理員，我希望每個月可以查看員工的點餐統計報表，以便分析營運狀況並作出採購和行銷決策。

- 作為餐廳管理員，我希望能查看所有訂單與評價記錄，以便即時了解服務品質並改善菜色或服務。

USER STORY-店員

- 作為店員，我希望能夠查看所有已下單的訂單列表（包含員工姓名、餐點內容與下單時間），以便掌握當前需處理的訂單。

- 作為店員，我希望能將每筆訂單的狀態從「待處理」更新為「製作中」、「已完成」、「已取餐」等狀態，以便讓員工與管理員即時掌握餐點進度。

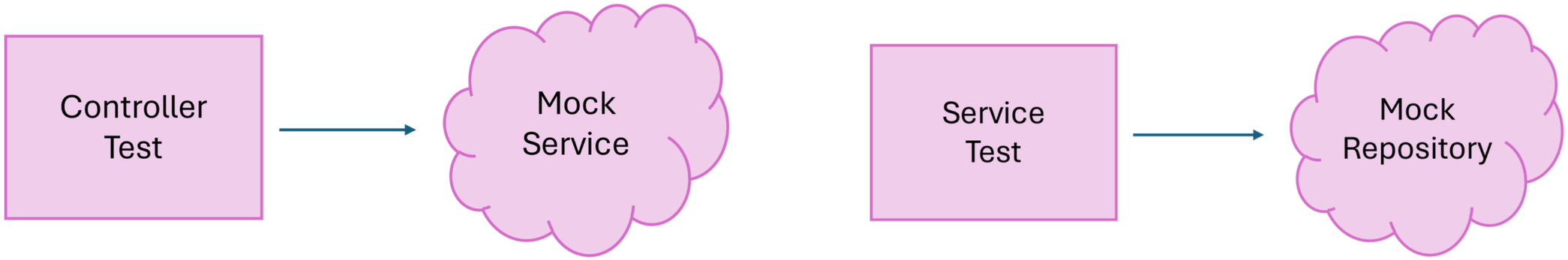
- 作為店員，我希望能依據訂單的狀態或時間排序訂單列表，以便優先處理即將到餐時間的訂單。

TEST

- Backend Unit Test
- load testing: k6

BACKEND UNIT TESTS

- Jest
- 分別測試每一個 Service 與 Controller



LOAD TEST-K6

```
(base) PS C:\Users\willi\Desktop\Codes\Project\Cloud Native\Cloud-Native-final>
>> k6 run loadtest.js

     /\      Grafana   /‾‾/
  /\  /  \     |\  __   /  /
 /  \/    \    | |/ /  /   ‾‾\
/          \   |   (  |  (‾)  |
/ __________ \  |_|\_\  \_____/

     execution: local
        script: loadtest.js
        output: -

     scenarios: (100.00%) 1 scenario, 300 max VUs, 4m20s max duration (incl. graceful stop):
              * default: Up to 300 looping VUs for 3m50s over 4 stages (gracefulRampDown: 30s, gracefulStop: 30s)


  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<1000' p(95)=247.12ms

    http_req_failed
    ✓ 'rate<0.01' rate=0.00%


  TOTAL RESULTS

    HTTP
    http_req_duration.....................................................: avg=230.85ms min=207.81ms med=229.48ms max=899.06ms p(90)=240.74ms p(95)=247.12ms
      { expected_response:true }...........................................: avg=230.85ms min=207.81ms med=229.48ms max=899.06ms p(90)=240.74ms p(95)=247.12ms
    http_req_failed.......................................................: 0.00%  0 out of 112491
    http_reqs.............................................................: 112491 488.1148/s

    EXECUTION
    iteration_duration....................................................: avg=894.98ms min=828.74ms med=891.53ms max=1.79s    p(90)=926.48ms p(95)=948.01ms
    iterations............................................................: 37497  162.704933/s
    vus...................................................................: 5      min=3
max=299
    vus_max...............................................................: 300    min=300        max=300

    NETWORK
    data_received.........................................................: 52 MB  226 kB/s
    data_sent.............................................................: 9.8 MB 43 kB/s

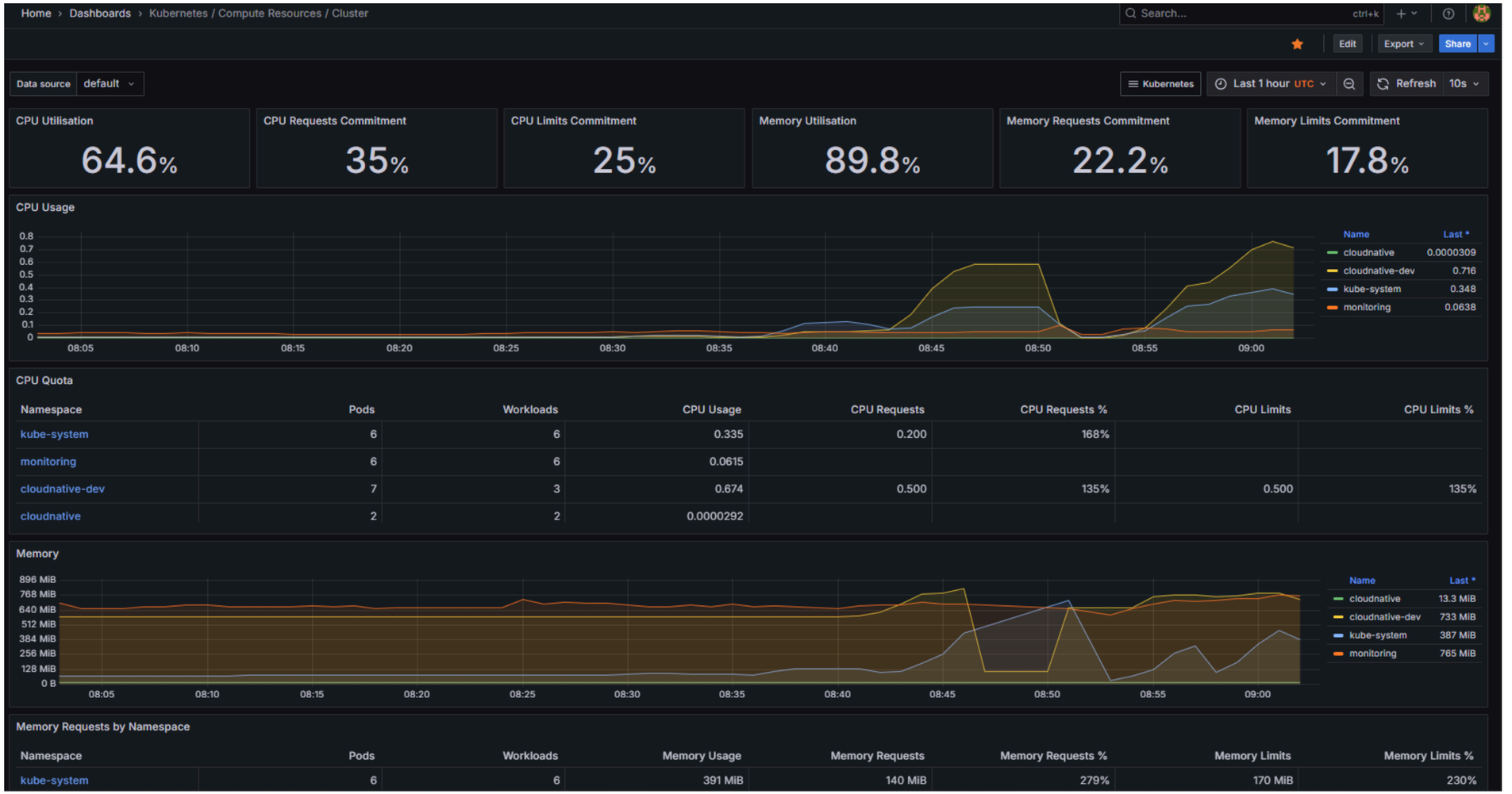


running (3m50.5s), 000/300 VUs, 37497 complete and 0 interrupted iterations
default ✓ [======================================] 000/300 VUs  3m50s
```

Success on

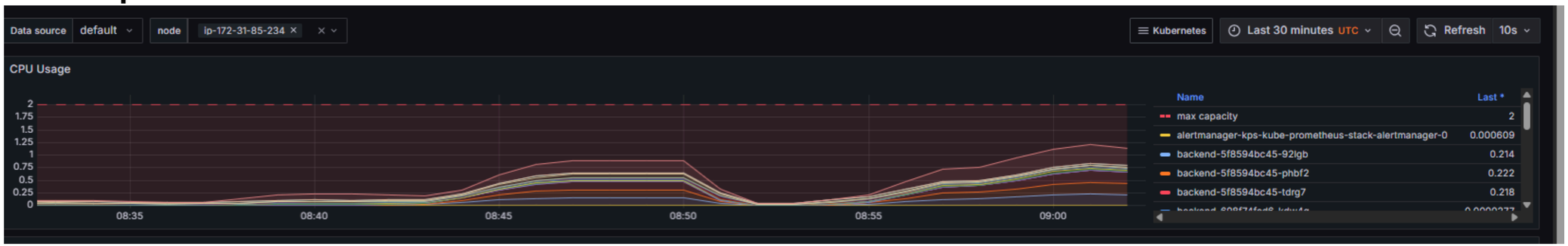
```
export const options = {
  stages: [
    { duration: '20s', target: 50 },    // 快速加壓，初始探測
    { duration: '1m',  target: 100 },   // 中壓穩定期
    { duration: '2m',  target: 300 },   // 高壓期：模擬尖峰
    { duration: '30s', target: 0 },     // 降壓
  ],
  thresholds: {
    http_req_failed: ['rate<0.01'],
    http_req_duration: ['p(95)<1000'],
  },
};
  // 每個使用者每 200ms 打完一輪請求
  sleep(0.2);
```

MONITOR-CLUSTER



MONITOR-POD

All pods



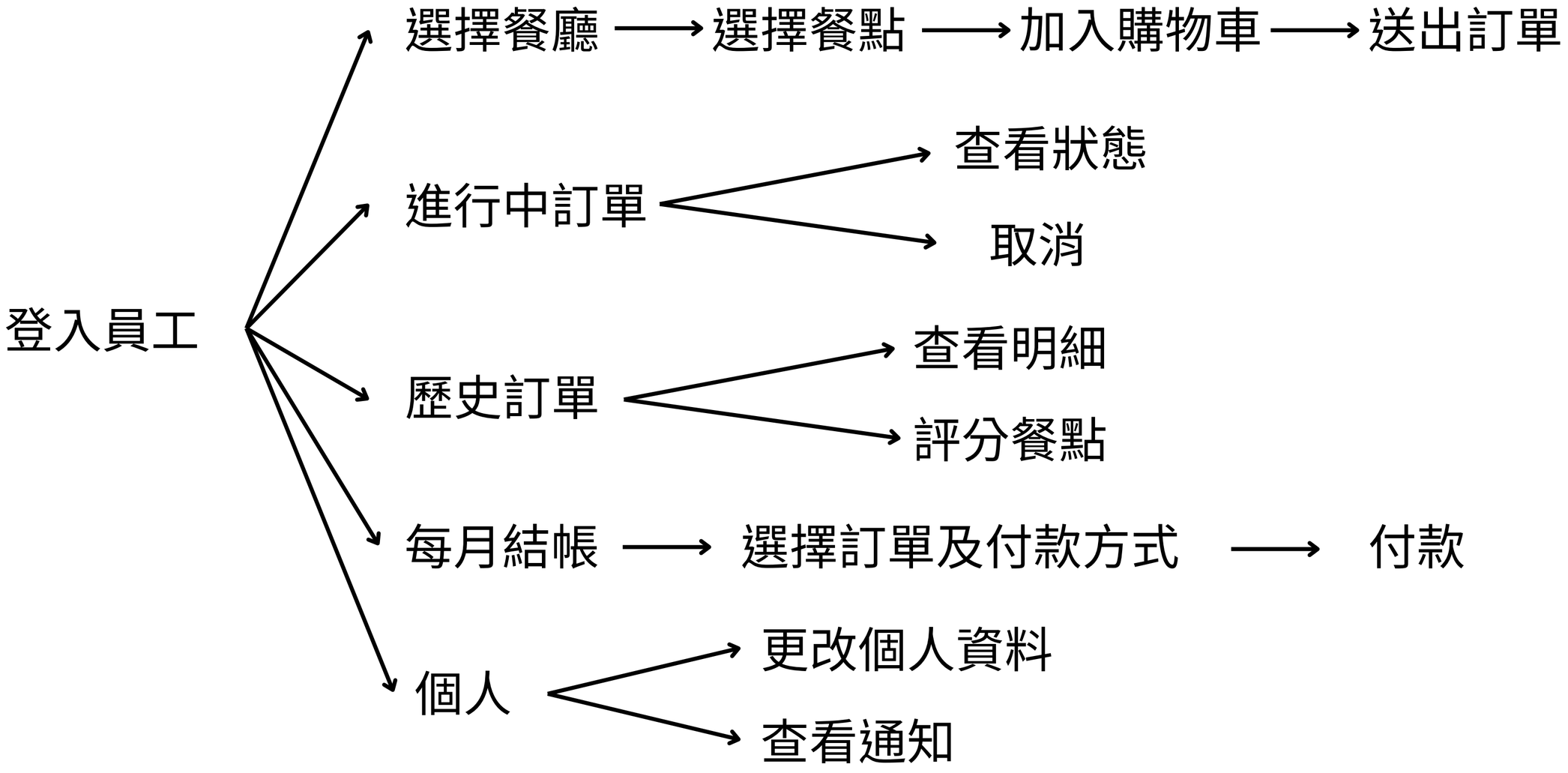
Single pod



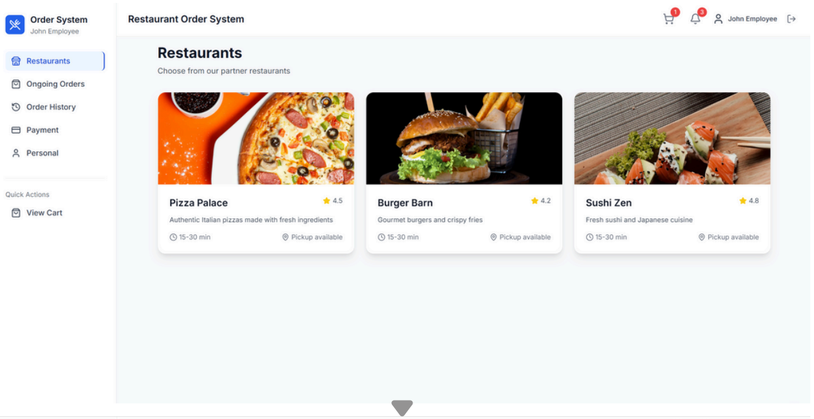
CICD Workflow

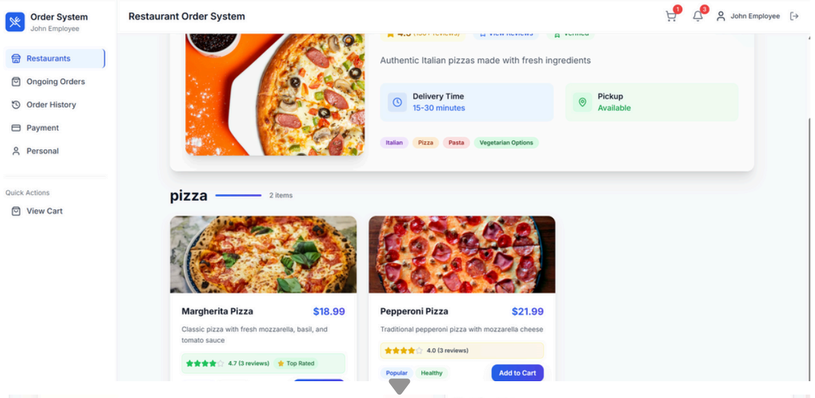
- 使用 GitHub Actions 建立 CI/CD 流程
- CI 執行 backend unit test
- docker build Frontend / Backend image
- SSH 遠端登入 EC2，部署至 K3s 叢集
- 自動等待 MySQL 就緒，確保後端啟動成功
- 支援多環境部署（main 分支為正式環境，develop 為 dev）

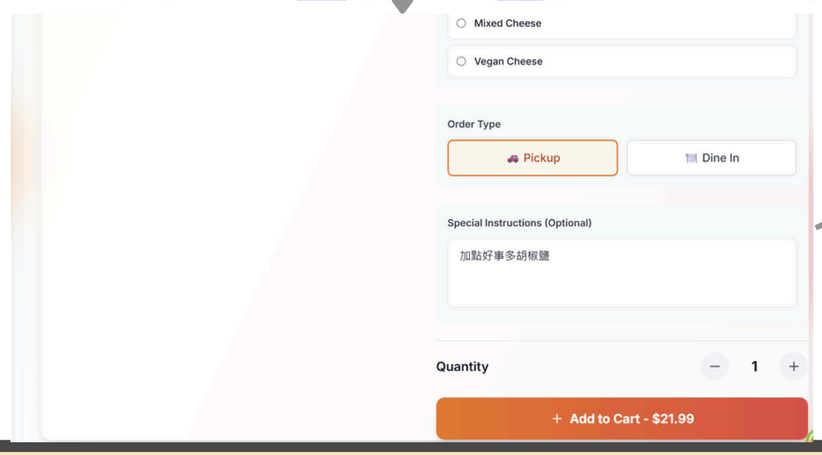
前端流程

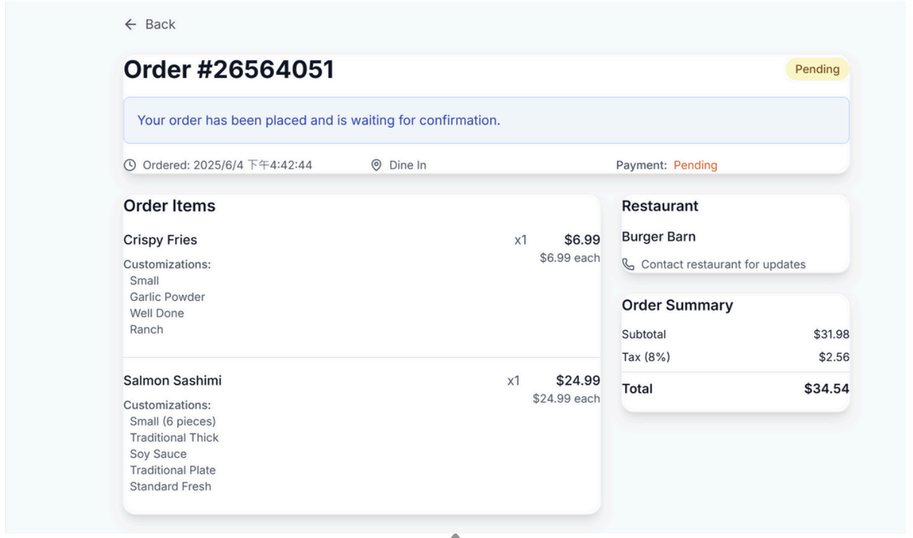

登入員工
選擇餐廳 → 選擇餐點 → 加入購物車 → 送出訂單
進行中訂單
查看狀態
取消
歷史訂單
查看明細
評分餐點
每月結帳 → 選擇訂單及付款方式 → 付款
個人
更改個人資料
查看通知

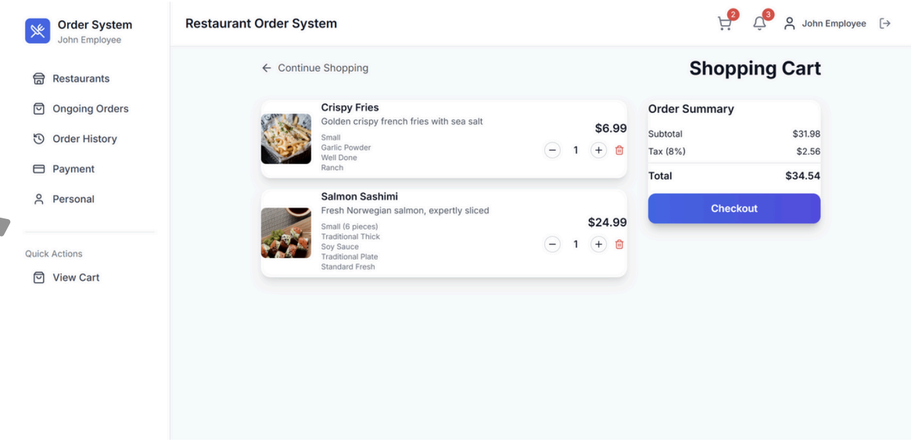
訂餐流程


Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Restaurants
Choose from our partner restaurants
Pizza Palace
4.5
Authentic Italian pizzas made with fresh ingredients
15-30 min
Pickup available
Burger Barn
4.2
Gourmet burgers and crispy fries
15-30 min
Pickup available
Sushi Zen
4.8
Fresh sushi and Japanese cuisine
15-30 min
Pickup available

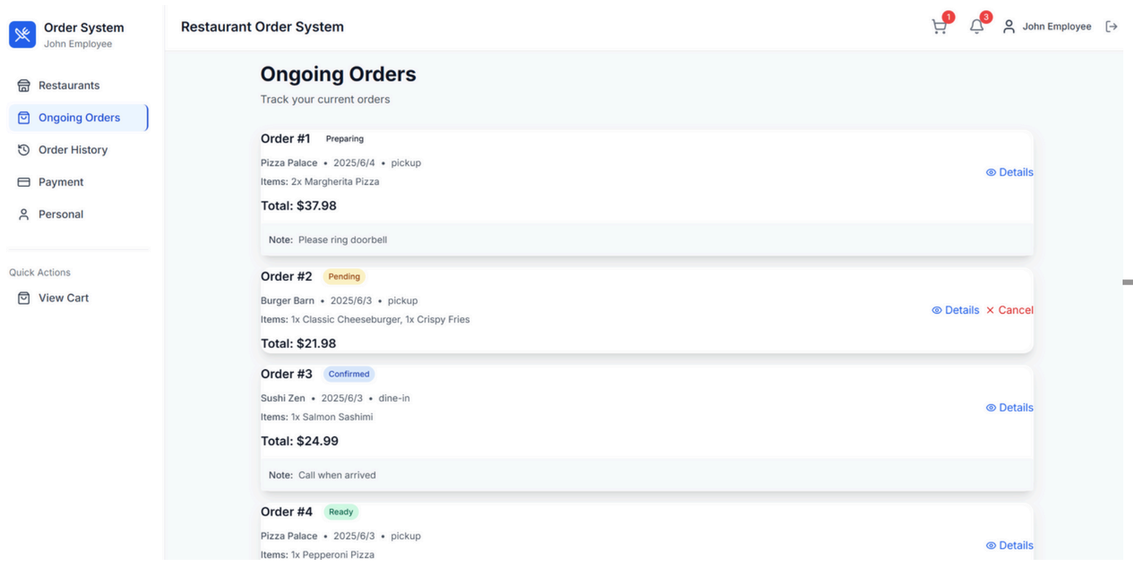

Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Authentic Italian pizzas made with fresh ingredients
Delivery Time
15-30 minutes
Pickup
Available
Italian
Pizza
Pasta
Vegetarian Options
pizza
2 items
Margherita Pizza
$18.99
Classic pizza with fresh mozzarella, basil, and tomato sauce
4.7 (3 reviews)
Top Rated
Pepperoni Pizza
$21.99
Traditional pepperoni pizza with mozzarella cheese
4.0 (3 reviews)
Popular
Healthy
Add to Cart

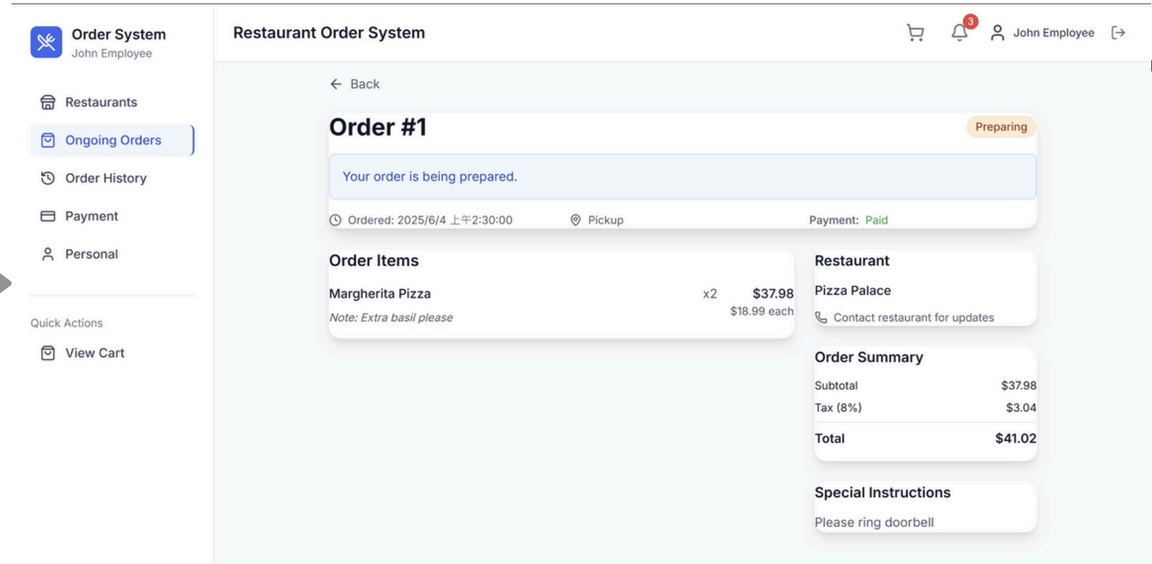

Mixed Cheese
Vegan Cheese
Order Type
Pickup
Dine In
Special Instructions (Optional)
加點好事多胡椒鹽
Quantity
1
Add to Cart - $21.99


Back
Order #26564051
Pending
Your order has been placed and is waiting for confirmation.
Ordered: 2025/6/4 下午4:42:44
Dine In
Payment: Pending
Order Items
Crispy Fries
x1
$6.99
$6.99 each
Customizations:
Small
Garlic Powder
Well Done
Ranch
Salmon Sashimi
x1
$24.99
$24.99 each
Customizations:
Small (6 pieces)
Traditional Thick
Soy Sauce
Traditional Plate
Standard Fresh
Restaurant
Burger Barn
Contact restaurant for updates
Order Summary
Subtotal
$31.98
Tax (8%)
$2.56
Total
$34.54

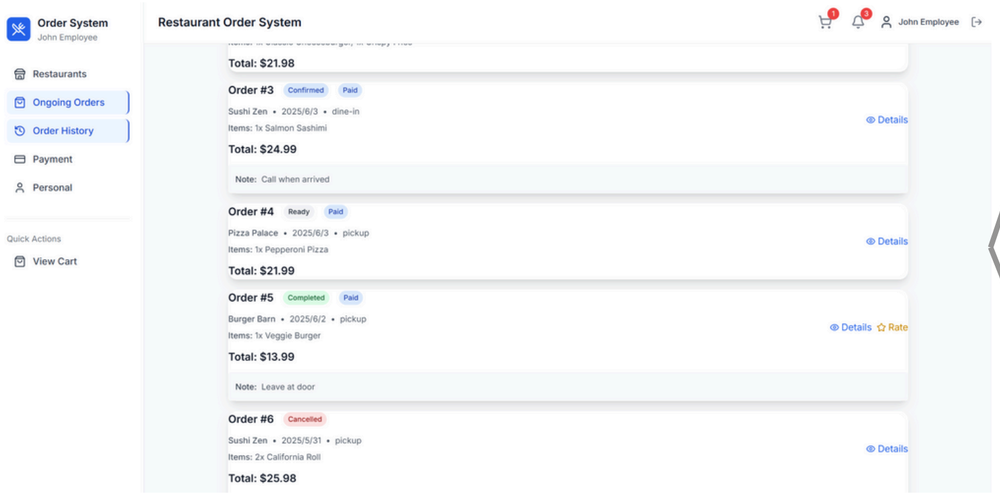

Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Continue Shopping
Shopping Cart
Crispy Fries
Golden crispy french fries with sea salt
Small
Garlic Powder
Well Done
Ranch
$6.99
1
Salmon Sashimi
Fresh Norwegian salmon, expertly sliced
Small (6 pieces)
Traditional Thick
Soy Sauce
Traditional Plate
Standard Fresh
$24.99
1
Order Summary
Subtotal
$31.98
Tax (8%)
$2.56
Total
$34.54
Checkout

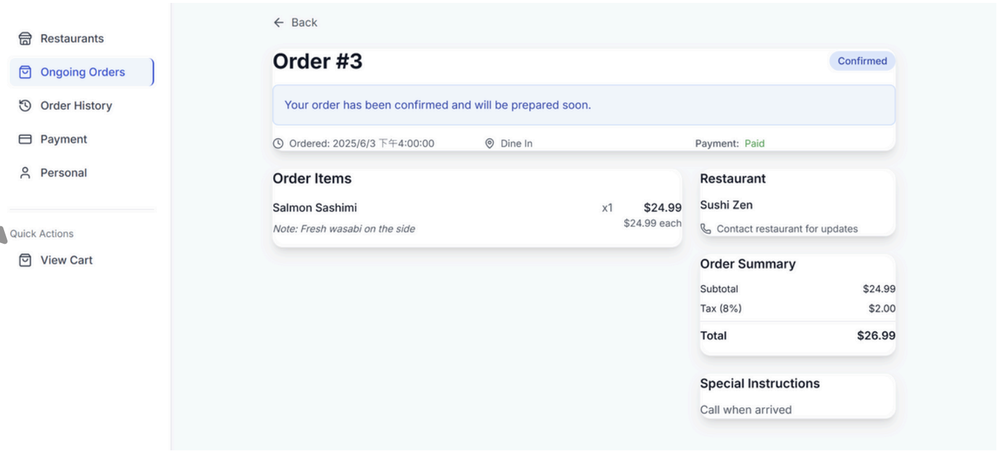
進行中訂單

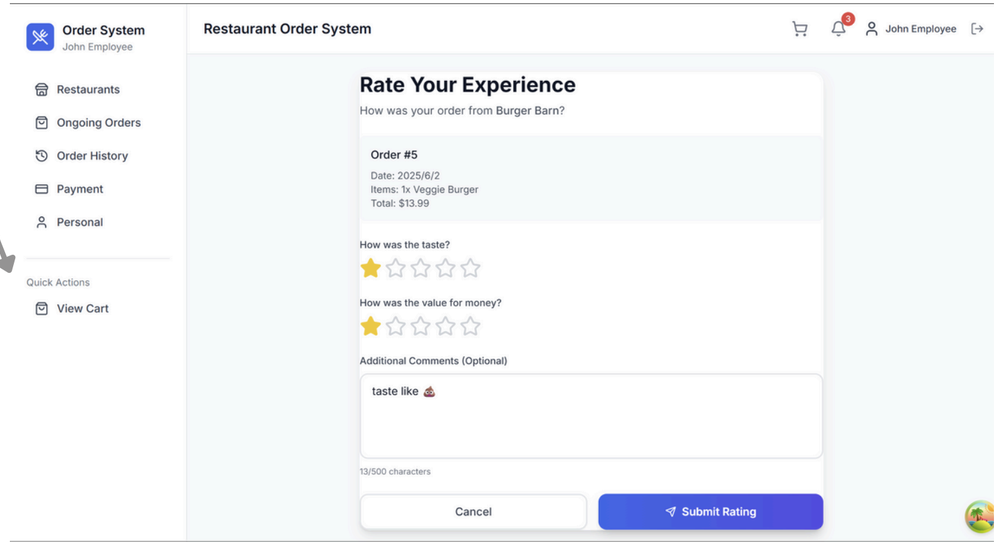

Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Ongoing Orders
Track your current orders
Order #1
Preparing
Pizza Palace • 2025/6/4 • pickup
Items: 2x Margherita Pizza
Details
Total: $37.98
Note: Please ring doorbell
Order #2
Pending
Burger Barn • 2025/6/3 • pickup
Items: 1x Classic Cheeseburger, 1x Crispy Fries
Details
Cancel
Total: $21.98
Order #3
Confirmed
Sushi Zen • 2025/6/3 • dine-in
Items: 1x Salmon Sashimi
Details
Total: $24.99
Note: Call when arrived
Order #4
Ready
Pizza Palace • 2025/6/3 • pickup
Items: 1x Pepperoni Pizza
Details


Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Back
Order #1
Preparing
Your order is being prepared.
Ordered: 2025/6/4 上午2:30:00
Pickup
Payment: Paid
Order Items
Margherita Pizza
x2
$37.98
$18.99 each
Note: Extra basil please
Restaurant
Pizza Palace
Contact restaurant for updates
Order Summary
Subtotal
$37.98
Tax (8%)
$3.04
Total
$41.02
Special Instructions
Please ring doorbell

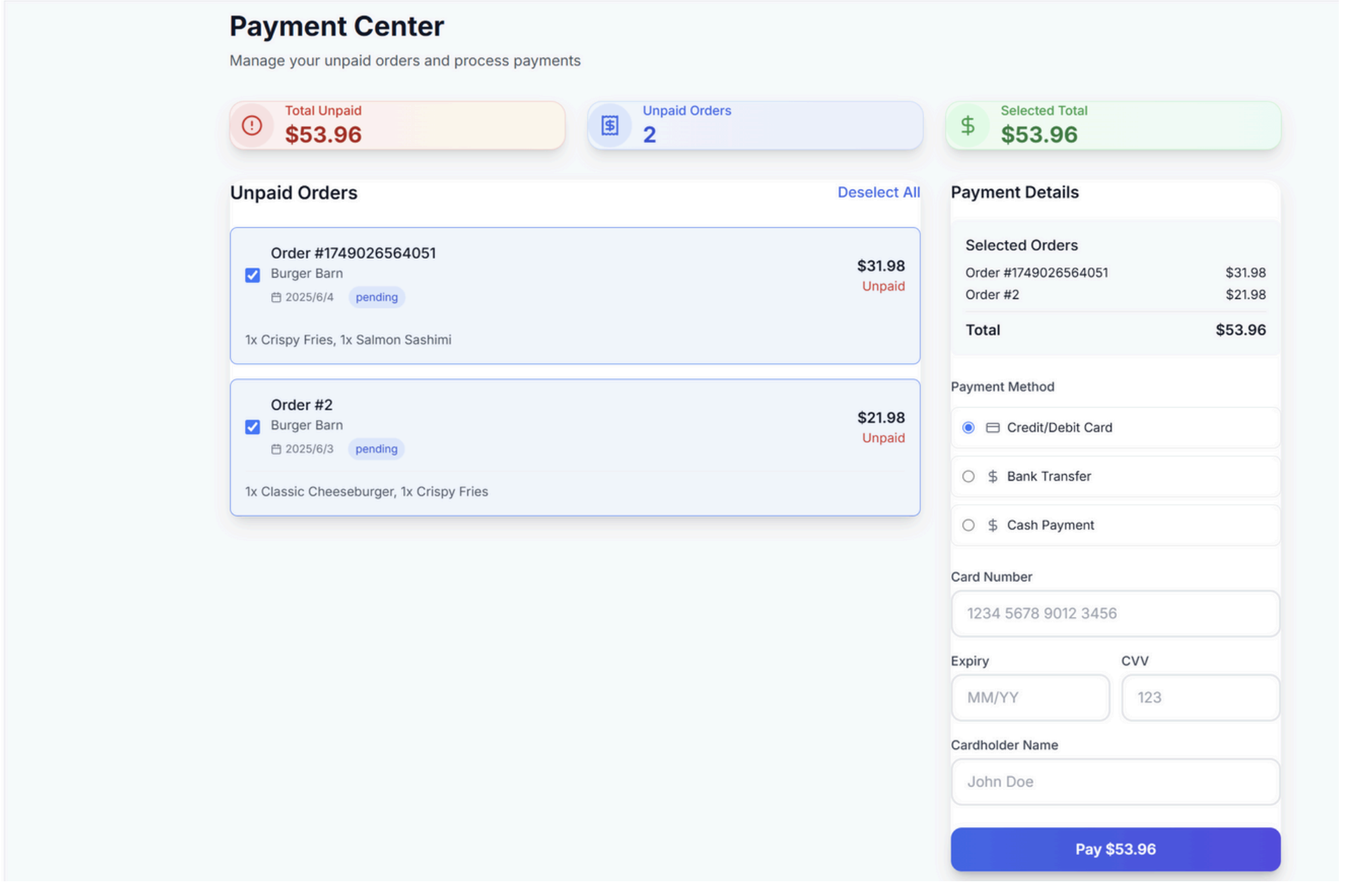
歷史訂單


Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Total: $21.98
Order #3 Confirmed Paid
Sushi Zen • 2025/6/3 • dine-in
Items: 1x Salmon Sashimi
Details
Total: $24.99
Note: Call when arrived
Order #4 Ready Paid
Pizza Palace • 2025/6/3 • pickup
Items: 1x Pepperoni Pizza
Details
Total: $21.99
Order #5 Completed Paid
Burger Barn • 2025/6/2 • pickup
Items: 1x Veggie Burger
Details Rate
Total: $13.99
Note: Leave at door
Order #6 Cancelled
Sushi Zen • 2025/5/31 • pickup
Items: 2x California Roll
Details
Total: $25.98

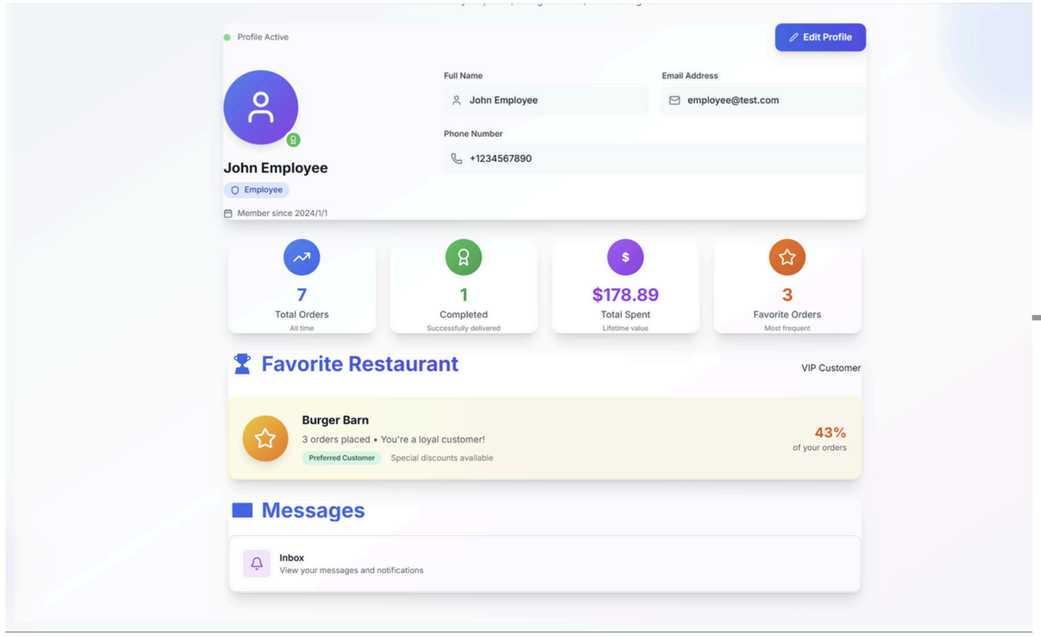

Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Back
Order #3
Confirmed
Your order has been confirmed and will be prepared soon.
Ordered: 2025/6/3 下午4:00:00
Dine In
Payment: Paid
Order Items
Salmon Sashimi
x1
$24.99
$24.99 each
Note: Fresh wasabi on the side
Restaurant
Sushi Zen
Contact restaurant for updates
Order Summary
Subtotal
$24.99
Tax (8%)
$2.00
Total
$26.99
Special Instructions
Call when arrived

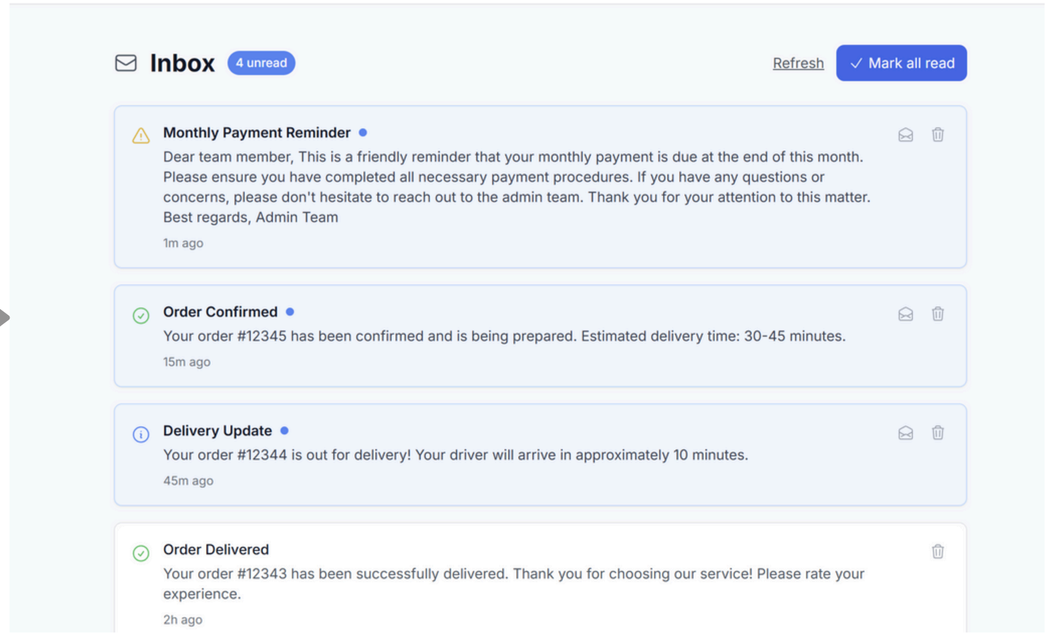

Order System
John Employee
Restaurants
Ongoing Orders
Order History
Payment
Personal
Quick Actions
View Cart
Restaurant Order System
John Employee
Rate Your Experience
How was your order from Burger Barn?
Order #5
Date: 2025/6/2
Items: 1x Veggie Burger
Total: $13.99
How was the taste?
How was the value for money?
Additional Comments (Optional)
taste like 💩
13/500 characters
Cancel
Submit Rating

付款

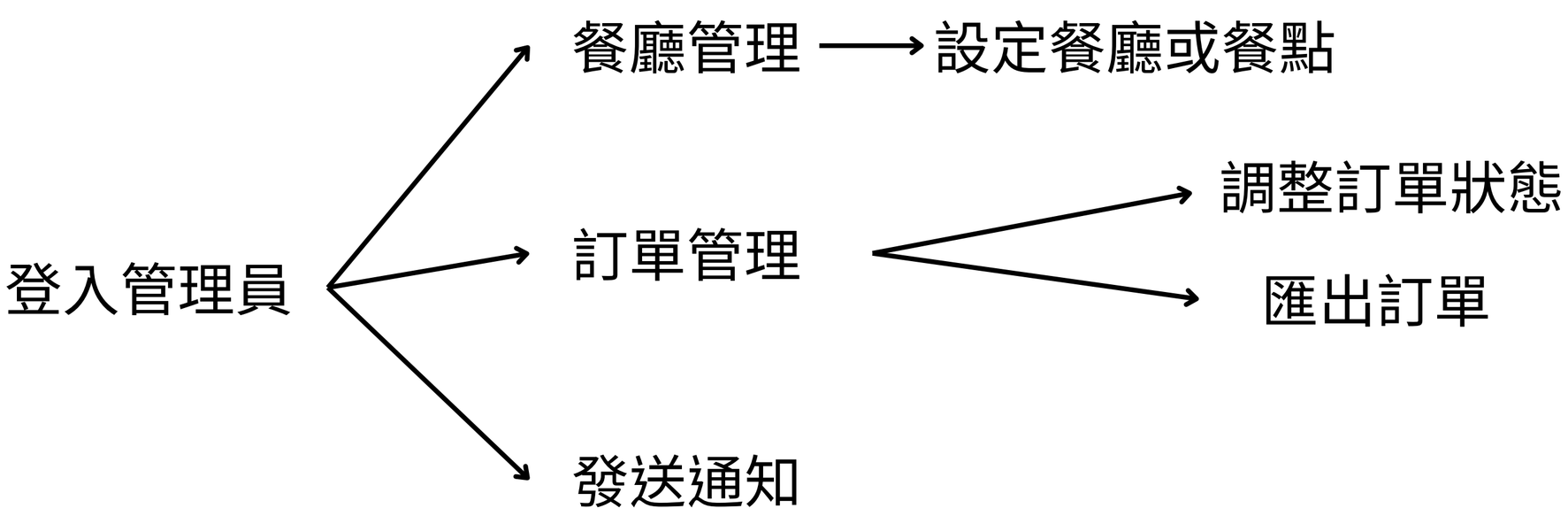

Payment Center
Manage your unpaid orders and process payments
Total Unpaid
$53.96
Unpaid Orders
2
Selected Total
$53.96
Unpaid Orders
Deselect All
Order #1749026564051
Burger Barn
2025/6/4
pending
$31.98
Unpaid
1x Crispy Fries, 1x Salmon Sashimi
Order #2
Burger Barn
2025/6/3
pending
$21.98
Unpaid
1x Classic Cheeseburger, 1x Crispy Fries
Payment Details
Selected Orders
Order #1749026564051 $31.98
Order #2 $21.98
Total $53.96
Payment Method
Credit/Debit Card
Bank Transfer
Cash Payment
Card Number
1234 5678 9012 3456
Expiry
MM/YY
CVV
123
Cardholder Name
John Doe
Pay $53.96

個人

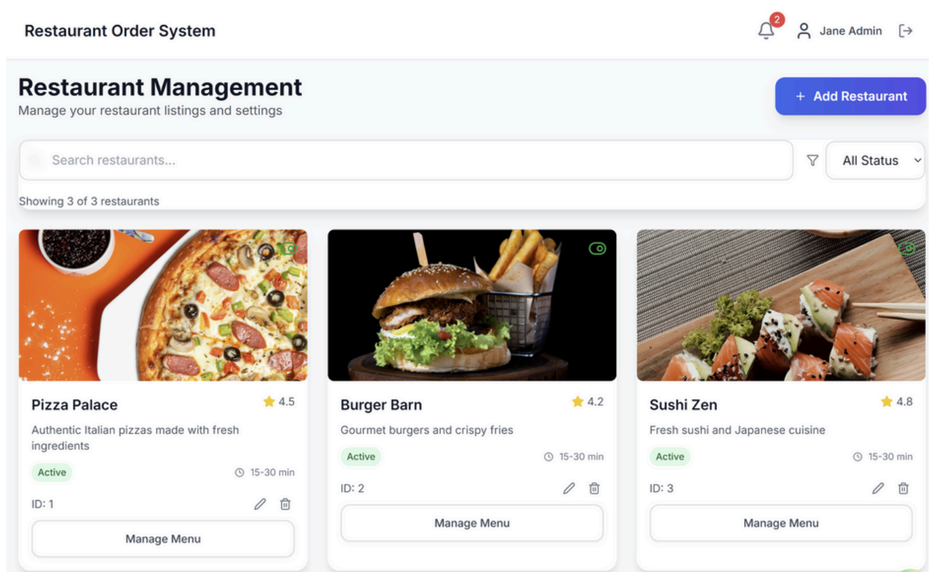

Profile Active
Edit Profile
John Employee
Employee
Member since 2024/1/1
Full Name
John Employee
Email Address
employee@test.com
Phone Number
+1234567890
7
Total Orders
All time
1
Completed
Successfully delivered
$178.89
Total Spent
Lifetime value
3
Favorite Orders
Most frequent
Favorite Restaurant
VIP Customer
Burger Barn
3 orders placed • You're a loyal customer!
Preferred Customer
Special discounts available
43%
of your orders
Messages
Inbox
View your messages and notifications

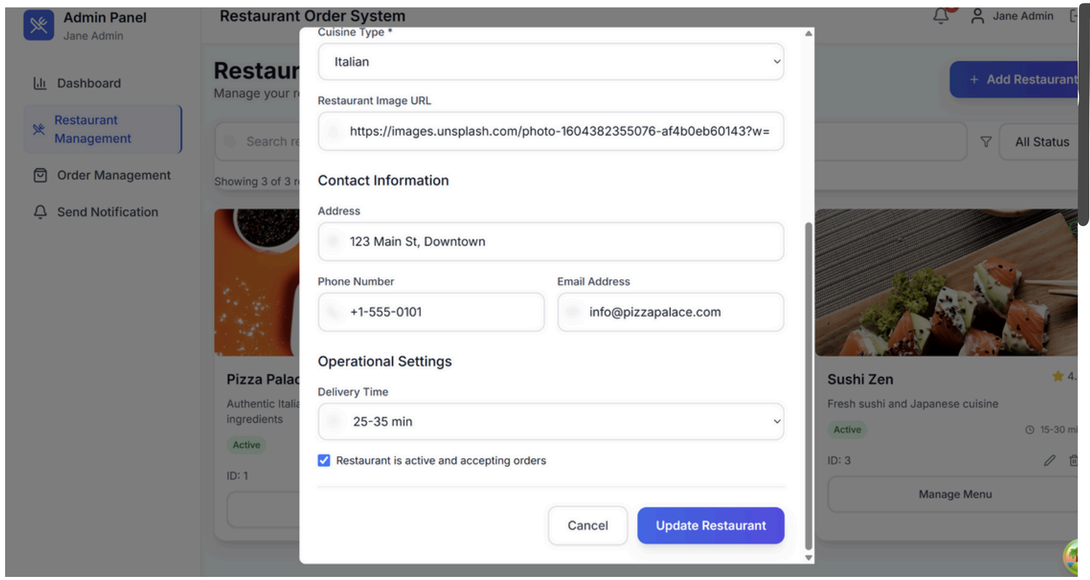

Inbox
4 unread
Refresh
Mark all read
Monthly Payment Reminder
Dear team member, This is a friendly reminder that your monthly payment is due at the end of this month. Please ensure you have completed all necessary payment procedures. If you have any questions or concerns, please don't hesitate to reach out to the admin team. Thank you for your attention to this matter. Best regards, Admin Team
1m ago
Order Confirmed
Your order #12345 has been confirmed and is being prepared. Estimated delivery time: 30-45 minutes.
15m ago
Delivery Update
Your order #12344 is out for delivery! Your driver will arrive in approximately 10 minutes.
45m ago
Order Delivered
Your order #12343 has been successfully delivered. Thank you for choosing our service! Please rate your experience.
2h ago

前端流程

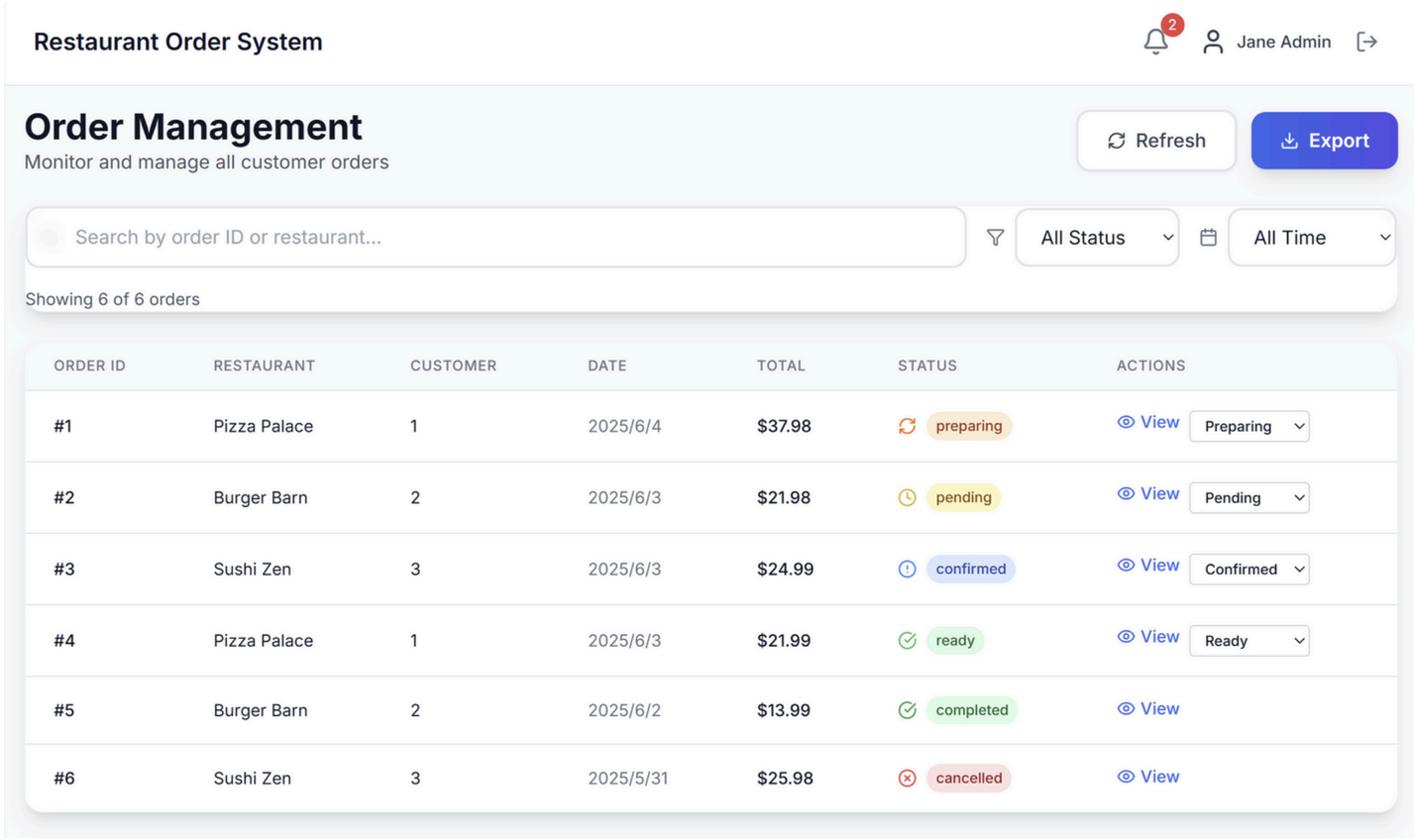


餐廳管理

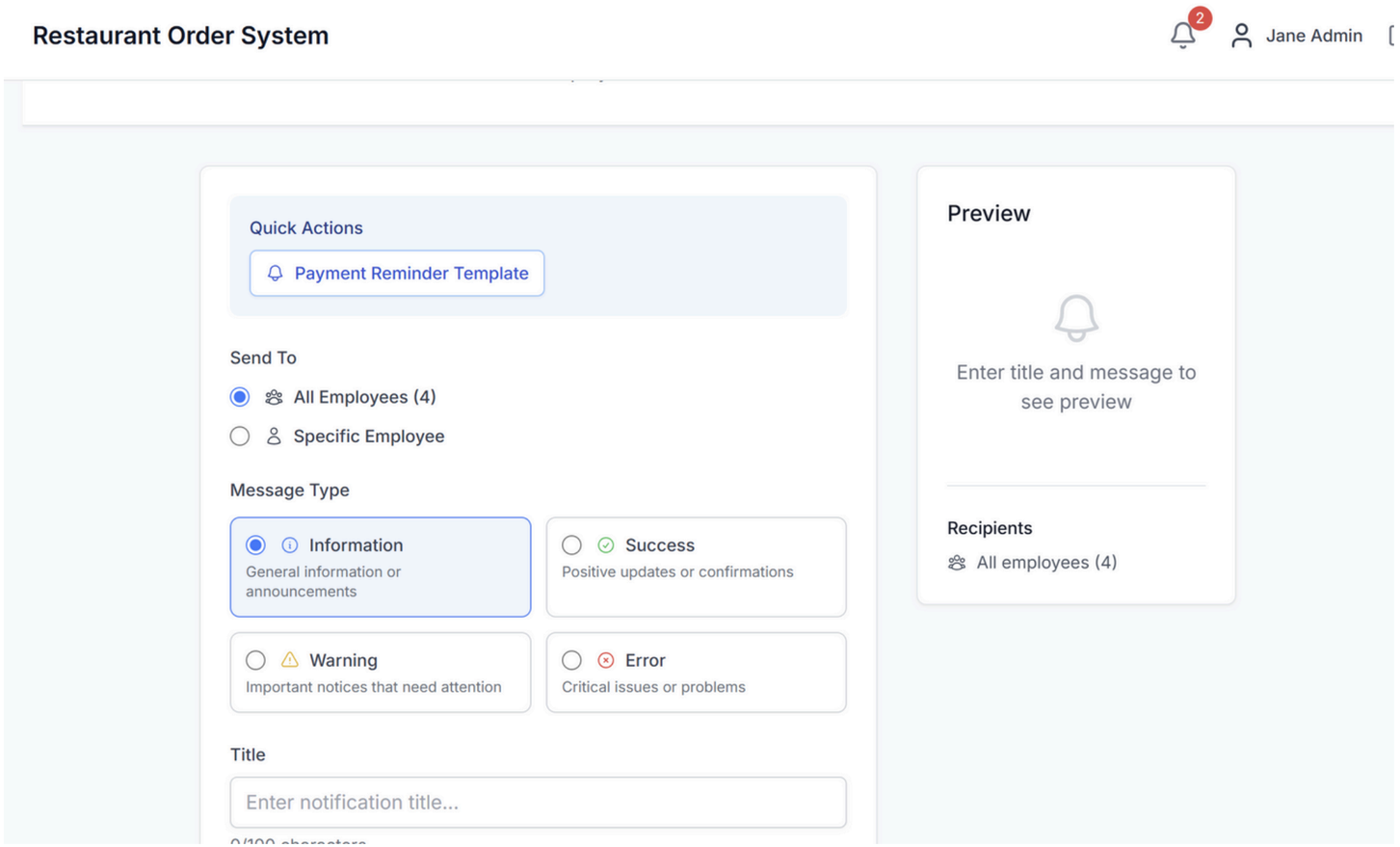

Restaurant Order System
Jane Admin
Restaurant Management
Manage your restaurant listings and settings
+ Add Restaurant
Search restaurants...
All Status
Showing 3 of 3 restaurants
Pizza Palace
4.5
Authentic Italian pizzas made with fresh ingredients
Active
15-30 min
ID: 1
Manage Menu
Burger Barn
4.2
Gourmet burgers and crispy fries
Active
15-30 min
ID: 2
Manage Menu
Sushi Zen
4.8
Fresh sushi and Japanese cuisine
Active
15-30 min
ID: 3
Manage Menu


Admin Panel
Jane Admin
Dashboard
Restaurant Management
Order Management
Send Notification
Restaurant Order System
Jane Admin
Cuisine Type *
Italian
Restaurant Image URL
https://images.unsplash.com/photo-1604382355076-af4b0eb60143?w=
Contact Information
Address
123 Main St, Downtown
Phone Number
+1-555-0101
Email Address
info@pizzapalace.com
Operational Settings
Delivery Time
25-35 min
Restaurant is active and accepting orders
Cancel
Update Restaurant

訂單管理


Restaurant Order System
2
Jane Admin
Order Management
Monitor and manage all customer orders
Refresh
Export
Search by order ID or restaurant...
All Status
All Time
Showing 6 of 6 orders
ORDER ID
RESTAURANT
CUSTOMER
DATE
TOTAL
STATUS
ACTIONS
#1
Pizza Palace
1
2025/6/4
$37.98
preparing
View
Preparing
#2
Burger Barn
2
2025/6/3
$21.98
pending
View
Pending
#3
Sushi Zen
3
2025/6/3
$24.99
confirmed
View
Confirmed
#4
Pizza Palace
1
2025/6/3
$21.99
ready
View
Ready
#5
Burger Barn
2
2025/6/2
$13.99
completed
View
#6
Sushi Zen
3
2025/5/31
$25.98
cancelled
View

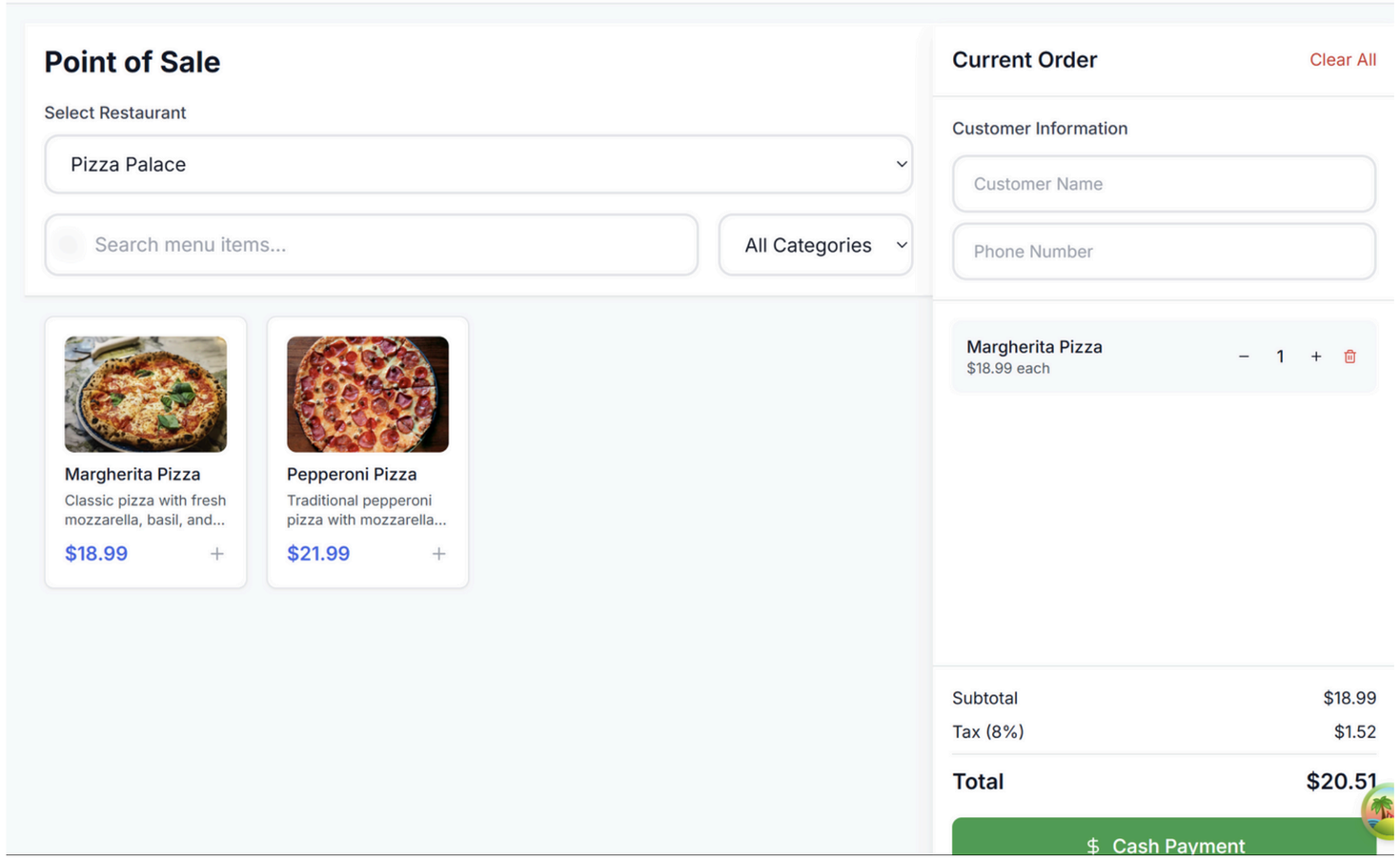
發送通知


Restaurant Order System
2
Jane Admin
Quick Actions
Payment Reminder Template
Send To
All Employees (4)
Specific Employee
Message Type
Information
General information or announcements
Success
Positive updates or confirmations
Warning
Important notices that need attention
Error
Critical issues or problems
Title
Enter notification title...
Preview
Enter title and message to see preview
Recipients
All employees (4)

餐廳員工流程



POS系統


Point of Sale
Select Restaurant
Pizza Palace
Search menu items...
All Categories
Margherita Pizza
Classic pizza with fresh mozzarella, basil, and...
$18.99
Pepperoni Pizza
Traditional pepperoni pizza with mozzarella...
$21.99
Current Order
Clear All
Customer Information
Customer Name
Phone Number
Margherita Pizza
$18.99 each
1
Subtotal
$18.99
Tax (8%)
$1.52
Total
$20.51
Cash Payment

LINK

http://13.218.27.133/
http://13.218.27.133/dev/frontend/

THANK YOU